GRADE 7 ACCELERATED · UNIT 4 · LESSON 5

Evaluating Expressions with Radicals

Question Bank 4.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{81}$

9

2. $\sqrt[3]{125}$

5

3. $\sqrt[3]{-8}$

−2

An odd root keeps the sign of its radicand.

4. $\sqrt{\frac{4}{25}}$

$\frac{2}{5}$

The numerator and the denominator each get the root.

5. $\sqrt{5^2+11}$

6

Finish under the bar: 25 + 11 = 36.

6. $\sqrt[3]{3^3+37}$

4

27 + 37 = 64.

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{100} \cdot 2^{-2}$

$\frac{5}{2}$

$10 \cdot \frac{1}{4}$.

b. $\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$

$\frac{17}{4}$

$\frac{1}{4} + 4$.

c. $\sqrt[3]{-64} - 3^{-1}$

$-\frac{13}{3}$

$-4 - \frac{1}{3}$.

8. Evaluate each expression.

a. $\sqrt[3]{27} - 1.4\left(\sqrt{3^2 - 5}\right)$

0.2

$3 - 1.4(2)$.

b. $\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$

$-\frac{17}{9}$

$\frac{1}{9} - 2$.

c. $\sqrt{4^3 - 39} \cdot 5^{-3}$

$\frac{1}{25}$

$5 \cdot \frac{1}{125}$.

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 12$

n = 144

b. $\sqrt[3]{n} = -5$

n = −125

c. $\sqrt{n+7} = 9$

n = 74

$n + 7 = 81$.

Reading someone else's work.

10. Tonya writes $\sqrt{16+9} = \sqrt{16} + \sqrt{9} = 7$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that her method ignores.

$\sqrt{16+9} = \sqrt{25} = 5$ ***. The bar groups 16 + 9 into a single radicand, so the addition is finished before any root is taken.***

Her method would need the root of a sum to equal the sum of the roots, and it does not.

11. Which is the value of $\sqrt{36+64}$?

- ○ A. 14
- ○ B. 100
- ● **C. 10**
- ○ D. 50

C

12. Select ALL of the expressions whose value is 3.

- ☐ A. $\sqrt{3^2+16}$
- ☒ **B.** $\sqrt[3]{2^2+23}$
- ☒ **C.** $\sqrt{4^2-7}$
- ☒ **D.** $\sqrt[3]{27}$
- ☐ E. $\sqrt[3]{-27}$
- ☒ **F.** $\sqrt{9}$

B, C, D and F

C's radicand is $16-7=9$ *and B's is* $4+23=27$.